Fragmentation outweighs heterogeneity in the accuracy cost of federated household load forecasting

Pushkal Gupta[1,*], Vanka Kaushik[1], Kalaavathi B[1]

[1] *School of Computer Science and Engineering, Vellore Institute of Technology, Vellore 632014, Tamil Nadu, India*

** Corresponding author: pushkal.gupta2023@vitstudent.ac.in*

E-mail addresses: vanka.kaushik2023@vitstudent.ac.in (V. Kaushik), kalaavathi.b@vit.ac.in (Kalaavathi B)

ORCID: 0009-0004-7032-6847 (Pushkal Gupta)

Abstract

Federated learning (FL) is proposed for household load forecasting because meter readings stay at home. Its accuracy loss is usually blamed on differences between households, and its bandwidth advantage is assumed. This study separated heterogeneity, data fragmentation and training budget on 100 London households from the Low Carbon London trial, with fixed homes, test period and long short-term memory forecaster. The homes were split over 1 to 100 clients in several ways, including a control that dealt training windows, not homes, to clients so that clients did not differ; heterogeneity was measured and every byte counted. With 30 rounds, the gap to centralised training grew from 2.0 % at five clients to 10.3 % with one home per client, and the control kept 78 % of that gap. Measured heterogeneity added a smaller, significant cost; over the tested ranges, fragmentation accounted for about four times as much error. Training for up to 100 rounds narrowed the gap at 100 clients to 7.4 % without closing it, and the pattern held in two further cohorts. Federated training moved more bytes than the raw readings in any encoding. With 20 clients of Dirichlet-mixed homes ($\alpha = 0.3$), federated averaging left 55 % of households more than 1 % worse off than local models, and local fine-tuning reduced this to 39 %. Without fine-tuning, federation was not significantly better than letting each client train alone on its pooled homes. Calendar inputs and proximal regularisation did not help. Federation here is a governance choice, not a bandwidth saving.

Keywords: smart meters; long short-term memory; non-IID data; communication efficiency; Low Carbon London; personalisation

1. Introduction

Short-term forecasts of single-household load feed demand response, home energy management, local flexibility markets and low-voltage network operation. Long short-term memory (LSTM) networks [1] forecast individual homes better than per-home statistical models when they are trained on many households at once [2], [3], but pooling the readings is exactly what smart-meter regulation and customer consent discourage, because half-hourly consumption reveals occupancy and daily routines.

Federated learning (FL) [4], [5] lets each meter, gateway or community hub train locally and send only model weights to an aggregator. Its use for residential load forecasting is growing [6]–[12], built on two premises: that household data are so heterogeneous (non-IID) that plain federated averaging (FedAvg) needs remedies such as FedProx [13], clustering [8], [14] or personalisation [9], [10], and that FL is the communication-light option [15], [16].

Neither premise has been tested under controlled conditions: studies rarely hold the households and test set fixed while varying only how the data are split. Three causes of the accuracy lost by FL are therefore entangled: heterogeneity between clients, the fragmentation of the data into many small clients, and the optimisation budget, since smaller clients take fewer local steps per round. Bandwidth claims rest on analytic estimates [15], [17]. This paper separates these factors on 100 homes of the Low Carbon

London (LCL) trial [18], with an independent and identically distributed (IID) control that fragments the data without making clients different, an extended training budget, and two further cohorts.

The principal contributions of this work are as follows:

- A controlled decomposition of the federated–centralised gap with an IID control and an extended budget, showing that most of the gap is a cost of fragmenting the data: clients that did not differ kept 78–83 % of it, while measured heterogeneity added a smaller, significant share and longer training narrowed but did not close it.
- A byte ledger of federated training against four encodings of the raw readings, showing that federation of a small LSTM moves more data than it keeps local, and that the traffic effect of heterogeneity is small and depends on the threshold used to measure it.
- A household-level analysis of who loses from federation, with a 1 % margin and a client-only baseline, showing that federation between community hubs pays off only with local fine-tuning, which also removes most of the harm to individual homes.
- Negative results for calendar inputs and for proximal regularisation, whose validation-tuned coefficient was zero.

2. Literature Review

2.1 Federated household load forecasting

Pooling neighbouring households reduces over-fitting in deep household forecasters [3]; whether federation preserves that benefit is disputed. On 100 LCL households, Briggs et al. [19] found plain FedAvg 7.6 % worse in RMSE than localised learning, with clustering reducing the shortfall to 4.7 % and local fine-tuning reversing it; in the same study their centralised model was 4.8 % worse than localised learning. Gholizadeh and Musilek [8] also found local learning best on average, whereas Fekri et al. [7] found FedAvg comparable to, or better than, per-meter models on 19 homes. Personalised and clustered variants [9], [10], [12], [14], [20] are almost always motivated by household heterogeneity.

2.2 Communication cost of federation

Communication efficiency was a founding motivation of FL [4], [21], but for load forecasting the evidence points both ways. Taïk and Cherkaoui [15] estimated analytically that FL saves 90–97 % of the network load of uploading data, and Savi and Olivadese [16] reported up to 50 times less exchanged data, whereas Briggs et al. [17] argued without measurement that weight vectors cost more than half-hourly readings on smart-meter links.

2.3 Non-IID data, local steps and proximal regularisation

Statistical heterogeneity slows and biases FedAvg [5], [22]. Remedies include proximal regularisation (FedProx [13]), control variates (SCAFFOLD [23]), adaptive server optimisers [24] and personalisation, and heterogeneity is usually induced with a Dirichlet split over class labels [25], which has no direct analogue for regression. A separate line of work on local stochastic gradient descent shows that, even with IID data, the error of averaged local updates depends on the number of local steps between averaging and on how many workers share the data [26]–[28]. Splitting a fixed data set over more clients changes exactly these quantities, so a growing federated gap need not indicate heterogeneity at all. The FedProx coefficient is usually tuned by hand, with adaptive variants based on the training loss [13] or a performance index [29].

2.4 Research gap

The closest studies used LCL [19], London Hydro [7], EPCOR [8] and Pecan Street [15] homes. None separates heterogeneity from fragmentation and budget while holding households and test set fixed, and none counts bytes during training against the raw-data volume.

3. Methodology

3.1 Problem formulation

Let $x_{i,t}$ be the energy (kWh) used by household i in half-hour t, standardised with statistics from that household's training period. From the last $L = 48$ readings the model predicts the next half-hour,

$$\hat{x}_{i,t+1} = f_w\left(x_{i,t-L+1}, \ldots, x_{i,t}\right) \tag{1}$$

where f_w is a single-layer LSTM with 64 hidden units and a linear output ($P = 17{,}217$ parameters, 68.9 kB in 32-bit floats), trained with mean squared error and Adam (learning rate 10^{-3}, batch 256). In the federated arms, K clients hold disjoint subsets of the training windows; in round r the server sends the global weights w_r to a set of clients, each trains for E local epochs, and the returning set S'_r is averaged:

$$w_{r+1} = \sum_{k \in S'_r} p_k \, w_k, p_k = \frac{n_k}{\sum_{j \in S'_r} n_j} \tag{2}$$

where n_k is the number of training windows of client k [4]. Clients are stateless: each local update starts a fresh Adam optimiser from w_r, so no optimiser moments are carried between rounds.

3.2 Data

The public LCL release [18] contains half-hourly consumption of 5,567 London households. For calendar year 2013, standard-tariff households with at least 95 % coverage were eligible (4,034 homes); the time-of-use group was excluded because it received dynamic prices that year. The main cohort of 100 homes was drawn with a fixed seed, stratified by quintile of mean load; two further cohorts were drawn the same way with other seeds and share only three or four homes with it. Invalid or implausible readings (above 10 kWh per half-hour) were set to missing, gaps of up to 2 h were interpolated, and windows touching longer gaps were dropped. The split was chronological for every home: training January–August, validation September–October and test November–December 2013 (1,160,952 training and 292,052 test windows in the main cohort). Scalers, profile clusters and the heterogeneity index used training data only, and every model-selection decision used validation data only. The data are public and anonymised.

3.3 Arms and partitions

Fig. 1 summarises the study. Persistence ($\hat{x}_{t+1} = x_t$) and seasonal naive ($\hat{x}_{t+1} = x_{t-47}$) forecasts served as references. Local-only training fitted one model per household and centralised training one model to the pooled windows of all households; both ran for at most 30 epochs with early stopping on validation MAE (patience 4). A client-only arm let each federated client train alone on the homes it holds, with the same early stopping. FedAvg followed Eq. (2); FedProx added $\frac{\mu}{2}\|w - w_r\|^2$ to each local loss [13]; and FT fine-tuned the final global model for one epoch on each household's own data, at no communication cost.

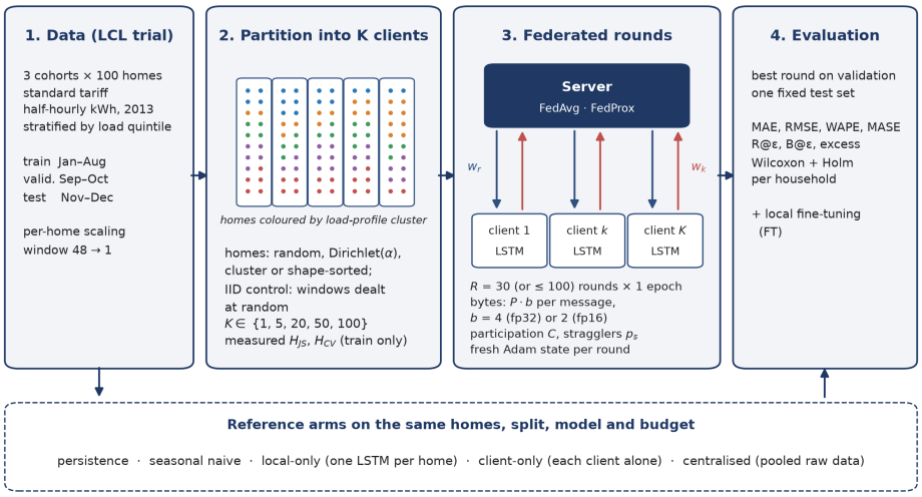


Figure 1. Study workflow. All arms used the same households, split and LSTM. Federated arms partitioned the training data into K clients and were tested once at the round chosen on validation error. w_r, w_k: global and client weights; $P \cdot b$: bytes per message; FT: local fine-tuning.

With one home per client, $K = 100$. Otherwise the same homes were grouped into K equal-sized clients of $100/K$ homes, as behind a building gateway or community energy hub. Each household was described by its normalised daily load shape and log mean load and clustered with k-means into $G = 5$ profile clusters. Five groupings of homes were used. *Random* grouping permuted the homes and cut the permutation into K blocks. *Dirichlet* grouping drew for each client a mixture $\pi \sim \mathrm{Dir}(\alpha \mathbf{1}_G)$ and filled each of its slots with a random home from a cluster drawn from π, renormalised over clusters with homes left [25]. *Cluster* grouping sorted homes by cluster and mean load and cut the list into blocks. *Shape* grouping sorted homes by the first principal component of their hour-of-week profile and cut the list into blocks, the largest difference in load shape that equal-sized clients of these homes allow. Finally, the *window* control was not a grouping of homes: the pooled training windows of all homes were dealt at random into K equal shards, so that every client saw the same mixture of homes. It fragments the data exactly as household grouping does while keeping clients statistically identical, so any gap it shows is due to fragmentation and budget, not heterogeneity. Validation and test errors were always computed per household on the same windows.

Measured heterogeneity. With q_k the distribution of client k's training consumption over the 168 hours of the week and JS the Jensen–Shannon divergence with base-2 logarithms [30], bounded in $[0, 1]$,

$$H_{\mathrm{JS}} = \binom{K}{2}^{-1} \sum_{k<l} \mathrm{JS}\,(q_k \| q_l) \tag{3}$$

H_{JS} measures differences in when clients consume. Differences in how much they consume were measured separately by the coefficient of variation of client mean load, H_{CV}; the two were not combined, because a composite would be dominated by the level term (it made up 94 % of the earlier composite index).

Communication. A message carries $P \cdot b$ bytes ($b = 4$ for 32-bit, 2 for 16-bit updates), with one download per selected client and one upload per returning client; framing headers are excluded. With v_r the validation MAE after round r and $v^\star$ that of centralised training, R@ε is the first round with $v_r \le (1+\varepsilon)v^\star$ and B@ε the bytes exchanged up to it, for $\varepsilon \in \{5{,}10{,}15{,}20\}\%$. Because any single threshold is arbitrary, the mean excess $\frac{1}{30}\sum_r (v_r/v^\star - 1)$ over the first 30 rounds was also reported. The raw-data volume was counted for four encodings: 2- and 4-byte readings, a 12-byte reading with a 4-byte timestamp offset and a 4-byte household index (the figure used for centralised traffic), and the

public comma-separated file. Stragglers (probability 0.2) completed a random fraction of their local work; FedAvg dropped them, FedProx kept their partial updates [13].

3.4 Experimental setup

The default federated setting used $K = 20$, full participation and $R = 30$ rounds of $E = 1$ epoch, the same number of passes over the data as the centralised budget. The experiments varied K (1, 5, 20, 50, 100), the grouping (window, random, $\alpha \in \{1, 0.3, 0.1\}$, cluster and shape), participation ($C = 0.5$ with stragglers), the aggregation rule and 16-bit updates. The local budget per round shrinks with K: a client took 4,534 Adam steps per round at $K = 1$, 227 at $K = 20$ and 46 at $K = 100$. FedAvg at $K = 20$, 50 and 100 was therefore rerun for up to 100 rounds, stopped after 25 rounds without a new best validation MAE. Two further checks addressed the forecaster: sine and cosine encodings of time of day and day of week as extra inputs, and the mean absolute scaled error (MASE), scaled per household by the in-sample one-step naive MAE on its training data. Each configuration ran with three seeds; the heterogeneity, participation and precision configurations ran with five. A seed fixed the model initialisation, the random grouping or Dirichlet draw, client sampling and stragglers, and the order of mini-batches; the cohort was fixed. The proximal coefficients were tuned on validation data over $\mu \in \{0, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$. The pipeline used Python 3.14.7, PyTorch 2.14.1 and an in-process simulator whose clients follow the Flower interface; Flower 1.39.0 [31] reproduced its results exactly.

Accuracy was measured by MAE, RMSE and weighted absolute percentage error (WAPE), micro-averaged over all test half-hours, and by MASE. For paired tests, each household's MAE was first averaged over seeds; arms were then compared with Wilcoxon signed-rank tests over the 100 households [32], with Holm correction [33] applied over the whole family of comparisons in Supplementary Table S1. A household was counted as harmed by an arm if its MAE was more than 1 % above its local-only MAE. Results with $p_{\text{Holm}} \geq 0.05$ are described as not significant, without implying equivalence.

4. Results

4.1 Main arms

Table 1 gives the test error of the main arms. Centralised training was the most accurate arm, better than persistence by 8.9 % in MAE, and better than local-only training for 72 % of households ($p_{\text{Holm}} < 10^{-4}$). Not every federated arm beat the naive references: with one home per client, FedAvg was slightly worse than persistence in MAE (0.0894 against 0.0890 kWh), although better in RMSE. As a sanity check of the federated implementation, running it on a single client holding all homes ($K = 1$) reproduced centralised accuracy ($p_{\text{Holm}} = 0.2092$).

4.2 Fragmentation and training budget

The gap to centralised training appeared as soon as the homes were split and grew with the number of clients (Fig. 2 and Table 2), from 2.0 % at $K = 5$ to 5.3 % at $K = 20$, 7.2 % at $K = 50$ and 10.3 % with one home per client. Under random grouping, however, splitting also makes clients more different: H_{JS} rose from 0.0028 at $K = 5$ to 0.0497 at $K = 100$, because a smaller client averages over fewer homes. The window control removed this confound. With windows dealt at random, H_{JS} stayed at or below 0.0094 (the residual is the sampling noise of finite shards) and client mean loads were equal, yet the gap remained 1.1, 4.3, 6.0 and 8.0 % at $K = 5$, 20, 50 and 100, that is 78–83 % of the gap with homes at $K \geq 20$.

Table 1. Test error of the main arms (kWh per half-hour; MAE as mean ± s.d. over seeds). Gap: MAE relative to centralised training. Traffic: upload plus download over 30 rounds; for centralised training, the raw training data at 12 bytes per reading. FT: after one epoch of local fine-tuning.

Arm	K	MAE	RMSE	WAPE (%)	Gap (%)	MAE, FT	Traffic (MB)
Persistence	–	0.0890	0.1987	37.6	–	–	0
Seasonal naive	–	0.1332	0.2773	56.3	–	–	0
Local-only	–	0.0840 ± 0.0003	0.1779	35.5	+3.6	–	0
Centralised	–	0.0811 ± 0.0003	0.1718	34.3	0.0	–	14.0
FedAvg, sanity check	1	0.0808 ± 0.0005	0.1710	34.2	−0.3	0.0812	4.1
FedAvg, random	5	0.0827 ± 0.0005	0.1721	35.0	+2.0	0.0818	20.7
FedAvg, random	20	0.0854 ± 0.0008	0.1758	36.1	+5.3	0.0833	82.6
FedAvg, random	50	0.0869 ± 0.0006	0.1782	36.8	+7.2	0.0842	206.6
FedAvg, one home per client	100	0.0894 ± 0.0013	0.1816	37.8	+10.3	0.0851	413.2
FedAvg, α=0.3	20	0.0860 ± 0.0003	0.1757	36.4	+6.0	0.0833	82.6
FedAvg, cluster	20	0.0857 ± 0.0003	0.1759	36.2	+5.7	0.0834	82.6
FedAvg, α=0.3, C=0.5, stragglers	20	0.0850 ± 0.0005	0.1759	36.0	+4.9	0.0833	37.2
FedAvg, α=0.3, 16-bit	20	0.0857 ± 0.0004	0.1755	36.3	+5.8	0.0833	41.3

A regression of run-level test MAE on $\ln K$ and H_{JS} over all FedAvg runs with random, Dirichlet, cluster and window partitions ($n = 54$, $R^2 = 0.92$) gave 1.37×10^{-3} kWh per unit of $\ln K$ (95 % CI 1.18 to 1.55, $p < 10^{-4}$) and 54.9×10^{-3} kWh per unit of H_{JS} (95 % CI 38.1 to 71.7, $p < 10^{-4}$). Both terms matter, but over the ranges tested going from five to 100 clients corresponds to 0.0041 kWh, whereas the whole range of H_{JS} obtained at $K = 20$ (0.0019 to 0.0213) corresponds to 0.0011 kWh.

Part of the fragmentation cost is a budget cost. With one local epoch per round, a client took 4,534 Adam steps per round at $K = 1$ but only 46 at $K = 100$, and the round selected on validation moved towards the end of the 30 rounds as K grew (Supplementary Table S2). Allowing up to 100 rounds narrowed the gap to 2.7 % at $K = 20$, 5.9 % at $K = 50$ and 7.4 % at $K = 100$. The $K = 50$ runs had stopped improving (best round 56, stopped at 76), but at $K = 20$ and $K = 100$ the best round was still near the end (96 and 95 of 100), so these gaps are upper bounds for longer training.

Table 2. Fragmentation with and without heterogeneity (FedAvg, 30 rounds). Homes: homes grouped at random; windows: training windows dealt at random to equal shards. Gap: test MAE relative to centralised training.

K	H_{JS}, homes	MAE, homes	Gap (%)	H_{JS}, windows	MAE, windows	Gap (%)
1	0.0000	0.0808	−0.3	–	–	–
5	0.0028	0.0827	+2.0	0.00050	0.0820	+1.1
20	0.0106	0.0854	+5.3	0.00194	0.0846	+4.3
50	0.0261	0.0869	+7.2	0.00477	0.0859	+6.0
100	0.0497	0.0894	+10.3	0.00939	0.0876	+8.0

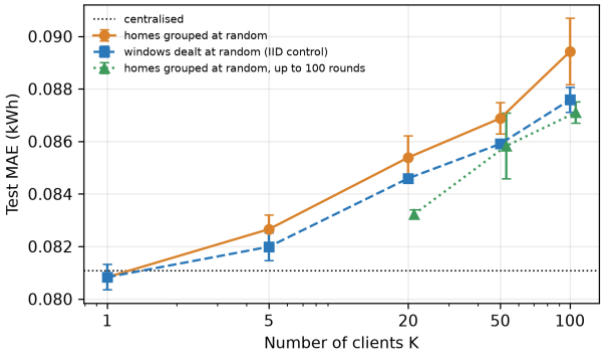


Figure 2. Test MAE (kWh) when the same 100 households are split over K clients (mean ± s.d. over seeds). Circles: homes grouped at random, 30 rounds; squares: training windows dealt at random (IID control), 30 rounds; triangles: homes grouped at random, up to 100 rounds. Dotted: centralised.

4.3 Heterogeneity at a fixed number of clients

At $K = 20$, grouping the homes from random to cluster almost doubled H_{JS} (Table 3), and cluster grouping also raised the spread of client mean load. Final test error hardly moved: across the groupings of homes it stayed within 0.0008 kWh, the 95 % intervals overlapped, and the rank correlation between H_{JS} and test MAE over runs was +0.14 ($p = 0.465$, $n = 30$). Per household, cluster grouping was not significantly different from random grouping ($p_{\mathrm{Holm}} = 0.0638$). The window control was the most accurate grouping (0.0846 kWh), consistent with the small heterogeneity term of the regression. Sorting homes by load shape gave the largest divergence ($H_{\mathrm{JS}} = 0.0213$) and the same error (0.0857 kWh), so the absence of an effect is not an artefact of the cluster definition; it does, however, hold only over the narrow range of heterogeneity that equal-sized clients of these homes allow.

The traffic needed to approach centralised accuracy depended more on the threshold and the seed than on the grouping (Fig. 3). Within 15 or 20 % of centralised validation error, every run of every grouping qualified after the first round (2.8 MB). Within 10 %, cluster grouping needed 48 % more traffic than random grouping on average over five seeds, but the IID control needed more than random grouping too (17.4 against 12.7 MB), so this difference is within seed variation. Within 5 %, many runs never qualified in 30 rounds. The threshold-free excess error over the first 30 rounds rose steadily but modestly, from 7.4 % for the window control and 8.5 % for random grouping to 9.3 %, so heterogeneity slowed FedAvg slightly. These traffic results concern FedAvg; the proximal variants converged differently (Supplementary Section S3).

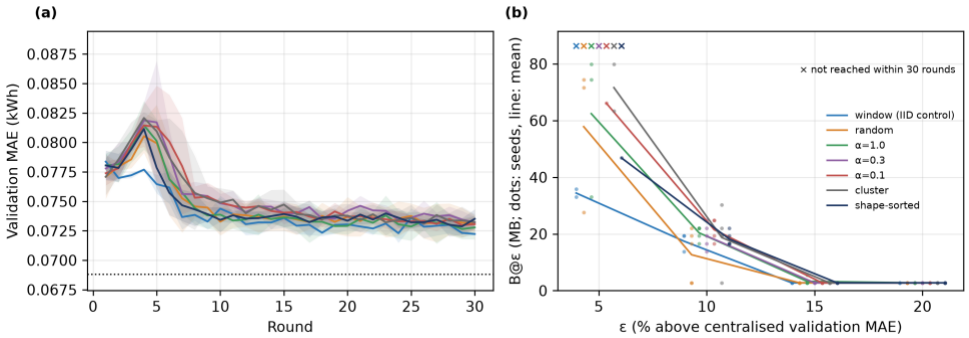


Figure 3. Groupings of the same homes into $K = 20$ clients (FedAvg). (a) Validation MAE per round (mean ± s.d. over seeds; dotted: centralised). (b) Megabytes exchanged before validation MAE first came within ε of centralised, for four thresholds; dots are seeds, crosses mark seeds that never reached the threshold.

Table 3. Groupings of the same homes into $K = 20$ clients (FedAvg). H_{JS}: load-shape divergence (base 2); H_{CV}: variation of client mean load; test MAE with 95 % confidence interval over seeds; B@ε: megabytes exchanged before validation MAE first came within ε of centralised, averaged over the seeds that reached it (in brackets); Excess: mean validation error above centralised over the first 30 rounds.

Grouping	Seeds	H_{JS}	H_{CV}	Test MAE (±95% CI)	B@5%	B@10%	B@20%	Excess (%)
window (IID control)	3	0.0019	0.006	0.0846 ± 0.0004	34.4 (2/3)	17.4 (3/3)	2.8 (3/3)	7.4
random	5	0.0106	0.320	0.0854 ± 0.0010	57.8 (3/5)	12.7 (5/5)	2.8 (5/5)	8.5
α=1.0	5	0.0125	0.333	0.0851 ± 0.0009	62.4 (3/5)	20.4 (5/5)	2.8 (5/5)	8.4
α=0.3	5	0.0152	0.301	0.0860 ± 0.0004	– (0/5)	19.3 (5/5)	2.8 (5/5)	9.5
α=0.1	5	0.0189	0.286	0.0856 ± 0.0005	66.1 (1/5)	21.5 (5/5)	2.8 (5/5)	9.3
cluster	5	0.0202	0.564	0.0857 ± 0.0004	71.6 (2/5)	18.7 (5/5)	2.8 (5/5)	9.3
shape-sorted	5	0.0213	0.350	0.0857 ± 0.0003	46.8 (1/5)	18.2 (5/5)	2.8 (5/5)	8.4

4.4 Forecaster strength, client-only training and other cohorts

Calendar inputs did not strengthen the forecaster (Supplementary Table S8): centralised MAE went from 0.0811 to 0.0815 kWh, and with one home per client FedAvg became worse (0.0919 kWh, a gap of 12.8 %). With MASE, centralised training (0.987) beat the in-sample naive scale while FedAvg with one home per client (1.111) did not; persistence scored 1.074 on the test period.

The client-only arm tests what federation adds once homes are pooled. Each of five clients training alone on its 20 homes reached 0.0836 kWh, against 0.0827 kWh for FedAvg over the same five clients; per household FedAvg was better for 54 % of homes, which was not significant ($p = 0.073$, uncorrected). At $K = 20$ (five homes per client) the two were again not significantly different (0.0849 against 0.0854 kWh, FedAvg better for 47 %, $p = 0.392$). Only with local fine-tuning did federation beat client-only training, for 66 % of households at $K = 5$ ($p < 10^{-4}$) and 63 % at $K = 20$ ($p = 0.013$).

The growth of the gap with K was repeated in the two further cohorts (Supplementary Table S5), although its size varied: at $K = 100$ it ranged from 3.8 to 10.7 % across the three cohorts, and at $K = 20$ from 0.9 to 5.7 %. The ranking of centralised and local training was not stable: local-only training was between -1.2 and 3.7 % from centralised, and in one cohort it was the better of the two.

4.5 Proximal regularisation

Validation tuning over a grid that included $\mu = 0$ selected $\mu = 0$, that is plain FedAvg (validation MAE 0.07238 against 0.07259 kWh for the best positive coefficient, $\mu = 0.001$; Supplementary Table S4). FedProx with that coefficient was practically indistinguishable from FedAvg (median paired difference -0.0001 kWh at $\alpha = 0.3$). An adaptive schedule driven by the disagreement between client updates did not improve on FedAvg either (Supplementary Section S3).

4.6 Communication

One message was 68.9 kB, so 30 rounds with $K = 20$ moved 82.6 MB and household clients moved 413.2 MB (Table 1). The eight months of training readings of the cohort (1.17 million values) occupy 2.3 MB at 2 bytes per reading, 4.7 MB at 4 bytes, 14.0 MB at 12 bytes and 58.1 MB in the public comma-separated format (about 50 bytes per row). Thirty rounds of federation at $K = 20$ thus moved more bytes than the readings in any of these encodings. Partial participation with stragglers lowered total traffic to 37.2 MB and 16-bit updates halved it (41.3 MB). Neither lost accuracy: 16-bit updates were marginally better than 32-bit for 92 % of households (median -0.0001 kWh, $p < 10^{-4}$) and partial participation for 94 % (median -0.0006 kWh, $p < 10^{-4}$). Both plausibly act as mild regularisers; the effects are below 1 % of MAE.

4.7 Who gains from federation?

Table 4 and Supplementary Fig. S2 give the per-household view. With the 1 % margin, centralised training harmed 24 % of households and helped 66 %. FedAvg at $K = 20$ harmed 45 % (random grouping) to 55 % ($\alpha = 0.3$) of households, and 75 % with one home per client. One epoch of fine-tuning reduced the share at $\alpha = 0.3$ to 39 %, after which FedAvg was not significantly different from local training (p_{Holm} = 0.4960); with one home per client it reduced the share to 51 %. The benefit of pooling was largest for the highest-consuming fifth of homes: their median MAE was 0.1393 kWh with local models, 0.1317 kWh with centralised training and 0.1408 kWh with FedAvg at $\alpha = 0.3$ (Supplementary Table S6).

Table 4. Share of households (%) whose test MAE was worse than local-only training at all, worse by more than 1 %, and better by more than 1 %. FedAvg at $K = 20$ unless stated; FT: after one epoch of local fine-tuning.

Arm	Worse than local (%)	Worse by >1% (%)	Better by >1% (%)
Centralised	28	24	66
FedAvg, random	56	45	31
FedAvg, α=0.3	66	55	28
FedAvg, α=0.3 + FT	49	39	42
FedAvg, K=100	82	75	14
FedAvg, K=100 + FT	61	51	26

5. Discussion

Most of the accuracy cost of federation in this setting came from fragmentation. Splitting the same data over more clients cost accuracy even when the clients were statistically identical, as described for local stochastic gradient descent: smaller shards give each client fewer steps between averages, and the average of many short local trajectories progresses less per round than one pass over the pooled data [26]–[28]. Part of this cost is recoverable with more rounds, at proportionally more traffic. Heterogeneity was a real but smaller cost, and it mainly slowed convergence. This is why drift remedies helped little here. The heterogeneity reachable with these homes was limited (H_{JS} at most 0.0497), and populations with photovoltaics, electric vehicles or heat pumps may differ much more, so the relative weight of the two costs should not be generalised beyond London households of this kind.

In practice, the number of clients was the most consequential choice. Federating five aggregated clients cost 2.0 % in MAE, meter-level federation 10.3 % with 30 rounds and 7.4 % with up to 100. Since one household's eight months of training readings occupy about 47 kB even at 4 bytes per value, less than one 68.9 kB message, federation at this scale is a governance choice based on data minimisation and consent, not a bandwidth saving. Operators should federate few, aggregated clients, budget rounds by local steps rather than epochs, fine-tune locally and check errors per consumption tier.

The results agree with Briggs et al. [19] that plain FedAvg can lose to local models on LCL, and that local fine-tuning recovers most of the loss. They differ on centralised training, which was the best arm here but 4.8 % worse than localised learning in their Table II. Their models were smaller (two layers of 20 units), used hourly sequences of 6 to 24 steps and trained for up to 500 epochs, and their centralised error was averaged over sequences rather than per client. The measured traffic supports the argument of Briggs et al. [17] rather than the analytic savings of Taïk and Cherkaoui [15], who compared a model of "1.9 kb" with 16 Mb of data. Their two LSTM layers of 200 units imply roughly 4.8×10^5 weights, about 1.9 MB in 32-bit floats, so the printed unit is probably a slip; with that correction their model would be of the same order as their data, as here.

Pooling homes into community clients requires a trusted holder, such as a building gateway, an energy community or a supplier that already reads the meters; an outside aggregator is not one. Where such a holder exists, the client-only results show that it gains little from federating with other hubs unless the shared model is fine-tuned locally. Where no intermediary is trusted, only the one-home-per-client setting applies, and there the fragmentation cost is largest.

The study covers one dataset and year, one horizon and a small LSTM. The forecaster is close to persistence (MASE near one), as is typical for single homes at this resolution, so absolute gains are small. The network was simulated, traffic counts payload only, and data locality is not a formal privacy guarantee [34], [35]: neither differential privacy nor secure aggregation was applied. Hyper-parameters other than the proximal coefficients and the number of rounds were fixed for all arms, and server-side adaptive optimisers [24] were not tested.

6. Conclusion

Holding households, test set and model fixed on 100 Low Carbon London homes and two further cohorts, this study found that the error of federated household load forecasting grew with the number of clients the same homes were spread over, to 10.3 % when every home was its own client, and most of that growth remained when clients received random shares of the data and did not differ at all. Measured heterogeneity added a smaller cost and slightly slowed convergence; training for longer narrowed but did not close the gap. Federated training moved more bytes than the readings it kept local, and one epoch of local fine-tuning removed most of the harm to individual households at community scale, while calendar inputs and proximal regularisation brought no benefit. The federated accuracy gap in this setting is mainly a fragmentation and budget cost, and the case for federation in residential forecasting rests on governance rather than bandwidth. Future work will test more heterogeneous populations, other horizons and larger models, and add differential privacy and secure aggregation.

Declarations

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Conflict of interest. The authors declare no conflict of interest.

Data availability. The Low Carbon London data are publicly available from the London Datastore (https://data.london.gov.uk/dataset/smartmeter-energy-consumption-data-in-london-households-vqm0d). All code, configuration files, per-run results and the scripts that generate every table and figure are available at https://github.com/Pushkal-Gupta/Short-Term-Household-Energy-Load-Forecasting.

Author contributions. Pushkal Gupta: conceptualisation, methodology, software, data curation, investigation, formal analysis, visualisation, writing – original draft. Vanka Kaushik: conceptualisation, investigation, validation, writing – review and editing. Kalaavathi B: supervision, methodology, writing – review and editing.

Declaration of generative AI use. During the preparation of this work the authors used Claude (Anthropic), an AI assistant, to help write and test the software pipeline and to draft and edit the text of the manuscript. After using this tool, the authors reviewed and edited the code, the results and the content as needed and take full responsibility for the content of the publication.

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